OS LAB COMPONENT:

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1. Program to work with chmod,chwon,permissions and its symbols:

Code:

```
echo "enter file name:"
read file
echo "enter permission (example 123):"
read perm
chmod $perm $file
echo "entere new owner name:"
read owner
sudo chown $owner $file
echo "new file permissions are"
ls -l $file
```

Output:

```
rvu@rvu-OptiPlex-SFF-7020:~/0093$ ./sa.sh
enter file name:
0093
enter permission (example 123):
456
entere new owner name:
rvu
new file permissions are
-r--r-xrw- 1 rvu rvu 0 Mar 10 14:13 0093
rvu@rvu-OptiPlex-SFF-7020:~/0093$
```

2.File permissions:

Code:

```
#!/bin/bash

usage()
{
    echo "Usage: $0 <filename>"
    exit 1
}

if [ "$#" -ne 1 ]; then
    usage
fi

filename=$1

if [ ! -e "$filename" ]; then
    echo "Error: File '$filename' does not exist."
    exit 1
fi

current_permissions=$(ls -l "$filename" | awk '{print $1}')
echo "Current permissions for '$filename': $current_permissions"

read -p "Enter new permissions (e.g., 755 for rwxr-xr-x): " new_permissions

if chmod "$new_permissions" "$filename"; then
    echo "Permissions for '$filename' changed to $new_permissions."
else
    echo "Error: Could not change permissions for '$filename'."
    exit 1
fi
```

```
new_permissions_display=$(ls -l "$filename" | awk '{print $1}')
echo "New permissions for '$filename': $new_permissions_display"

read -p "Do you want to change the ownership of the file? (y/n): " change_ownership

if [[ "$change_ownership" =~ ^[Yy]$ ]]; then
    read -p "Enter new owner (username): " new_owner
    read -p "Enter new group (groupname): " new_group

    if chown "$new_owner":"$new_group" "$filename"; then
        echo "Ownership for '$filename' changed to $new_owner:$new_group."
    else
        echo "Error: Could not change ownership for '$filename'."
        exit 1
    fi
else
    echo "Ownership change skipped."
fi

echo "Final details for '$filename':"
ls -l "$filename"
```

Output:

```
rvu@rvu-OptiPlex-SFF-7020:~/0093$ ./worm.sh copy.sh
Current permissions for 'copy.sh': -r--r-xr-x
Enter new permissions (e.g., 755 for rwxr-xr-x): 455
Permissions for 'copy.sh' changed to 455.
New permissions for 'copy.sh': -r--r-xr-x
Do you want to change the ownership of the file? (y/n): y
Enter new owner (username): rvu
Enter new group (groupname):
Ownership for 'copy.sh' changed to rvu:.
Final details for 'copy.sh':
-r--r-xr-x 1 rvu rvu 229 Feb 24 14:50 copy.sh
rvu@rvu-OptiPlex-SFF-7020:~/0093$
```